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3D Contour Generation based on Diffusion Probabilistic Models

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***Abstract*—The contours of objects effectively represent essential 3D information such as shapes and boundaries. Existing contour detection methods mostly rely on thresholding or neural networks to classify points as edge or non-edge points. However, these methods often lack generalization ability on datasets with different shapes, leading to issues such as missing contours and discontinuous distribution. To address this, we propose a 3D point cloud contour generation method based on the denoising diffusion probabilistic model (DDPM). Our method treats the complete point cloud as an explicit condition to guide the generation of contour from noise. Specifically, our method trains the DDPM to be a conditional generative network customized for contour generation tasks. In the network, we design the Conditional Feature Extraction (CFE) module that obtains multi-scale feature information, and the Conditional Feature Fusion (CFF) module embeds this information in the generation process to guide contour generation. The experimental results demonstrate the effectiveness of our method. The source code of our method is available at: *https://github.com/djzgroup/ContourGeneration*.**

***Index Terms*—point cloud, contour generation, diffusion model, generative AI**

I. Introduction

With the advancement of 3D sensors, 3D point clouds have become crucial for capturing 3D information due to their easy acquisition and high storage efficiency. Contours on object surfaces provide vital visual information and aid in tasks like classification and segmentation. Extracting object contours enables more accurate object localization and recognition, reducing data processing workload. However, representing shape contours in 3D point cloud data remains challenging. This paper focuses on generating point cloud contours from complete point cloud representations.

Detecting 3D contours in unstructured point cloud data, such as sharp features, feature contours, or curves, is a challenging task. Traditional methods [1], [2] involve surface mesh reconstruction, graph-based approaches, or local neighborhood analysis using principal component analysis. With the rapid development of machine learning, learning-based methods [3] have also been proposed to classify points as contour or non-contour. However, these methods are mostly suitable for objects with distinct geometric features, and the resulting contour points often suffer from missing or discontinuous distribution. In this paper, we explore a learning-based approach to generate point cloud contours.

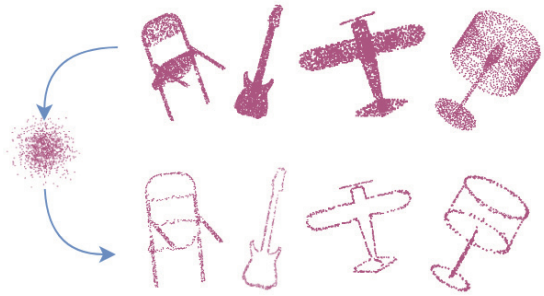

Fig. 1. The proposed method generates 3D contour from noise guided by the complete point cloud based on the diffusion probabilistic model.

Learning-based generative models have shown remarkable capabilities in representation learning for point clouds, enabling tasks like shape generation, completion, upsampling, and synthesis. However, these models, such as VAEs, GANs, and flow-based models, which have achieved success in the 2D image domain [4], [5], cannot be directly applied to point clouds due to the irregular and unordered distribution of points in three-dimensional space. Therefore, learning generative models for point clouds is highly challenging. Subsequently, there is increasing research has explored methods like GANs [6], auto-regressive models [7], and flow-based models [8] for generating point clouds. Despite significant progress, these methods have inherent limitations in point cloud generation modeling. For instance, GANs' training process can be unstable due to adversarial loss, and auto-regressive models assume an unnatural generation ordering, limiting model flexibility.

Compared to these works, denoising diffusion probabilistic models (DDPM) [9] can generate uniform and high-quality point clouds. It iteratively moves a set of Gaussian noise towards a complete and clean point cloud, defining a one-to-one pointwise mapping between two consecutive point clouds during the diffusion process, allowing it to be trained using a simple and efficient mean squared error loss function that requires the generated point cloud to be uniform.

Hence, the task of 3D contour generation can be treated as a point cloud generation problem [10] using DDPM, and

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the shape of the complete point cloud can be treated as a guiding condition for the generation. However, previous works have focused on generating complete point clouds, and have not shown direct effectiveness in contour generation tasks, thus lacking specialized network architectures for the task. Therefore, we propose a conditional point cloud contour generation method based on the diffusion model, as shown in Fig. 1. Our method regards the complete point cloud as the guidance to generate the corresponding point cloud contour from noise. Our main contributions can be summarized as follows:

- A 3D point cloud contour generation method is proposed, which trains a denoising diffusion model as a conditional generative network customized for our task.
- The point cloud features extracted from the complete shape are adopted as the guiding condition in the process of contour generation to enhance accuracy.
- A feature extract module is designed to extract multi-scale features, and a feature fusion module to fuse conditional features with denoising point clouds.
- Experiment results show that the proposed network achieves competitive performance, as well as good generalization across different datasets.

II. Related work

A. Contour detection

Contour and 3D edges extraction in point clouds can be achieved through various techniques. Traditional methods for contour detection typically rely on the geometric information of 3D objects, such as sharp features, depth data variation, and surface mesh reconstruction, to extract edge points. Che et al. [2] analyzed data distribution based on the normal variation to predict contour points. Monga et al. [11] focused on 3D edge detection of point clouds using recursive filtering. Edelsbrunner [1] detected boundaries via alpha-shapes and DBSCAN-based 3D detection methods. However, these methods still rely on appropriately setting decision thresholds, requiring extensive experimentation, and have low generalization ability for contour detection tasks on different shapes.

With the rapid development of deep learning, learning-based methods have been proposed to classify individual points as contour or non-contour points. Contour points classification has been achieved by using multi-level feature information and contour classifiers [3], [12]. Pie-net [13] trains two classifiers using the PointNet++ [14] architecture to classify points into corner points and edge points. Additionally, Loizou et al. [15] proposed graph convolution-based part contour detection in point clouds, further advancing the development of learning-based neural networks focused on contour detection. Although using deep neural networks for contour detection has achieved certain results, issues of inaccurate and discontinuous contour points still exist.

B. Point cloud generation

Regarding the generation models for point clouds, previous research has explored various methods such as Variational Autoencoders (VAEs) [16] and Generative Adversarial Networks (GANs) [17]. These methods can generate point clouds with a fixed number of points and directly rely on common loss functions in point clouds, such as Earth Mover's Distance (EMD) and Chamfer Distance (CD). The Snowflakenet [18] network has shown good performance in point cloud generation through using Skip-Transformer and Snowflake point deconvolution. Additionally, the viewpoint of considering point clouds as point distributions has inspired the application of likelihood-based methods in point cloud generation. Pointflow [8] employs continuous normalizing flows to model the distribution of points. PointGrow [7] is a self-autoregressive model that assumes inputs are ordered and have precise likelihood. The existing point cloud generation methods have shown good performance and versatility in point cloud generation, but their performance in contour generation is not ideal.

Our proposed method of contour generation is related to the diffusion probabilistic model [9]. The diffusion probabilistic model learns the distribution of the data by using a Markov chain to transform noise into data that approximates the real distribution. It iteratively moves a set of Gaussian noise to a clean point cloud. DDPM [9] defines a one-to-one mapping between two point clouds during the diffusion process, which allows for efficient training using a simple mean squared error loss function. In contrast, generative models based on GANs [6], [17] are difficult to train stably due to complex adversarial loss, flow-based methods [8] involve expensive ODE integrations, and autoregressive models [7] require data to be processed in a specific order, lacking flexibility. Our method is diffusion-based conditional generation, with a simple objective function for training and high-quality generation.

III. Method

The point cloud contour generation task can be seen as a conditional generation problem based on DDPM, where the complete point cloud serves as a condition. We propose a diffusion-based condition point cloud contour generation method that generates the corresponding contour guided by the condition from the complete point cloud. As shown in Fig. 2, our proposed network first utilizes a Conditional Feature Extraction (CFE) module to extract contour features from the complete point cloud as a condition. Subsequently, a Conditional Feature Fusion (CFF) module incorporates the feature into the denoise process, guiding the generation of continuous and effective point cloud contours.

A. Problem statement

To generate reasonable and effective point cloud contours, our training objective is to maximize the similarity between the generated contours and the real contours, i.e., to maximize the log-likelihood probability of conditional generation. In our proposed method G_θ (θ represents the model parameters), we assume that the training dataset D consists of M pairs, $D = \{(X_j, Y_j)\}, 1 \leqslant j \leqslant M$, the Y_j denotes a complete point cloud data, where X_j denotes the corresponding point cloud contour. A point cloud object is represented by N points in

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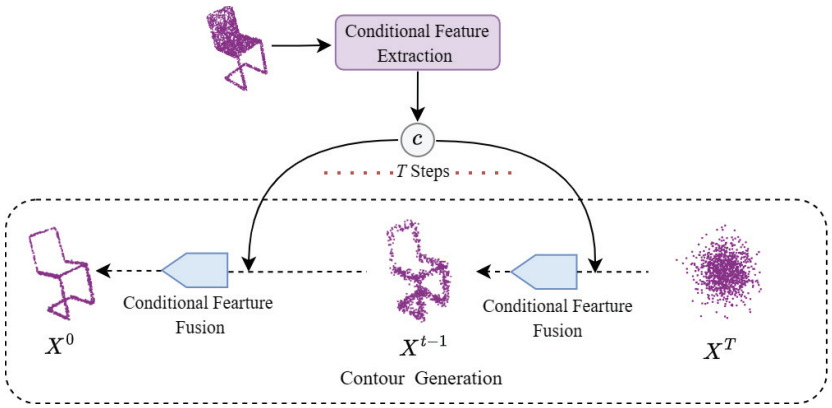

Fig. 2. The pipeline of our diffusion-based conditional point cloud contour generation network. The proposed network trains the denoising diffusion probabilistic model as a generative neural network for contour generation, under the guidance of a complete point cloud.

3D space: $X = \{x_i | 1 \leqslant i \leqslant N\}$, each of which $x_i \in R^3$ is the coordinate of the i-th point. Features c_j is extracted from the complete point cloud data Y_j. The parameters of the generated network are θ, and p_{data} is the distribution of point cloud contour x_i in the generation process, with T generation steps. Therefore, the training objective of maximum likelihood estimation can be expressed as:

$$argmaxlogpG_\theta(X_j|c_j) \quad j = 1, 2, ..., M. \tag{1}$$

B. Denoising diffusion process

Our contour generation is based on the denoising diffusion process with condition c. The denoising diffusion process is divided into diffusion and denoise processes as follows:

The diffusion process gradually adds Gaussian noise to the clean data distribution p_{data} until the output approximates Gaussian noise p_{noise}. Dhariwal et al. [19] provides mathematical proof that the diffusion process is independent of the contour feature c of the complete point cloud. Formally, let $x^0 \sim p_{data}$, for conciseness, we omit the subscription i in the following discussion. The diffusion process from clean data x^0 to x^t is defined as:

$$q(x^1, \cdots, x^T \mid x^0) = \prod_{t=1}^{T} q(x^t \mid x^{t-1}), \tag{2}$$

where $q(x^t \mid x^{t-1}) = \mathcal{N}(x^t; \sqrt{1-\beta_t}x^{t-1}, \beta_t \boldsymbol{I})$. The hyper-parameters β_t are predefined small positive constants. We first define the constants $\alpha_t = 1 - \beta_t, \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$, and then, the x^t can be directly sampled by the following equation, the parameter ϵ is the standard Gaussian noise.

$$x^t = \sqrt{\bar{\alpha}_t}x^0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}. \tag{3}$$

The denoise process predicts and eliminates the Gaussian noise added in the diffusion process. During this process, our proposed CFF module guides contour generation, depends on the obtained conditional feature c. Let $x^T \sim p_{noise}$, The process from x^T to clean data x^0 is defined as:

$$p_\theta(x^0, \cdots, x^{T-1} \mid x^T, c) = \prod_{t=1}^{T} p_\theta(x^{t-1} \mid x^t, c), \tag{4}$$

where $p_\theta(x^{t-1} \mid x^t, c) = \mathcal{N}(x^{t-1}; \mu_\theta(x^t, t, c), \sigma_t^2 I)$. The mean value $\mu_\theta(x^t, t, c)$ is parameterized by a neural network with θ, and the variance σ is a constant related to the time step. To generate the corresponding contours, we sample the complete point cloud Y, and encode it with the proposed CFE module to obtain c. Guided by the condition c, sample x^T, through the CFF module, and then draw the $x^{t-1} \sim p_\theta(x^{t-1} \mid x^t, c)$ for $t = T, T-1, \cdots, 1$, and finally output x^0.

It can be seen that the parameterized network to be learned in the denoise process is $\mu_\theta(x^t, t, c)$, and $\mu_\theta(x^t, t, c) = \frac{1}{\sqrt{\alpha_t}}\left(x^t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(x^t, t, c)\right)$, where ϵ_θ is a neural network that takes the noisy point cloud $x^t \sim q(x^t \mid x^0)$, generation step t and conditional features c as inputs. Finally, the simplified loss function is formalized as:

$$L(\theta) = \mathbb{E}_{i\sim N, t\sim T} \parallel \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t}x_i^0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t, c) \parallel^2 \tag{5}$$

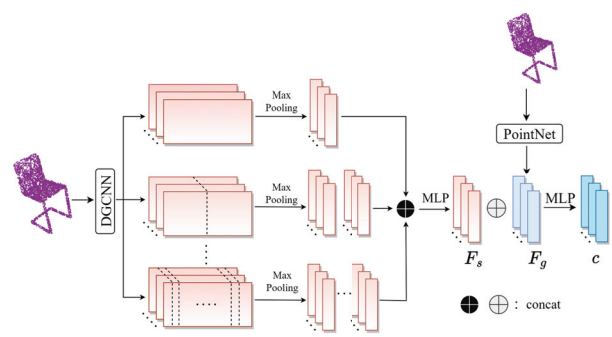

Fig. 3. Conditional Feature Extraction (CFE) module. The module extracts multi-scale local contour features and concatenates a global feature to obtain conditional feature c as guidance in contour generation.

C. Conditional point cloud contour generation network

In our proposed method G_θ, the reasonable contour X is generated by the denoising diffusion process. The conditional feature c, which guides the generation, is obtained from the complete point cloud Y through the Conditional Feature Extraction module. Then, the feature c is propagated to the denoise process through the Conditional Feature Fusion

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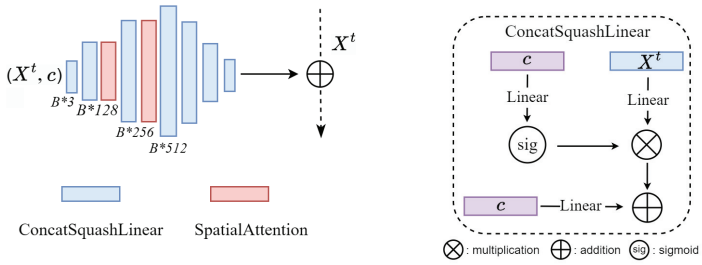

(a) Condition Feature Fusion module (b) ConcatSquashLinear layer

Fig. 4. Condition Feature Fusion module: (a) The CCF module consists of a series of ConcatSquashLinear layers and SpatialAttentions layers. ConcatSquashLinear integrates condition c into X^t and then adds SpatialAttention to emphasize the edge information of the point cloud. (b) Conditional feature c and X^t are transformed into the same dimension via linear layers, followed by matrix multiplication, and finally added by the transformed conditional feature c.

module, as shown in Fig. 2. The network takes noisy point cloud X^T, complete point cloud Y, and generation step size T as inputs. During the sampling generation process, the network iteratively outputs $X^{t-1}, ..., X^0$ based on the time step T, resulting in clean contours.

Conditional Feature Extraction module: Based on the extraction of point-wise features in point clouds, DGCNN [20] effectively captures local features of point clouds through dynamic graph convolution, making it an outstanding point cloud feature extractor. In our proposed network, the Conditional Feature Extraction module is designed to extract the conditional feature c for the contour generation. As shown in Fig. 3, the module first obtains per-point features based on DGCNN from the complete point cloud. To minimize the information loss and obtain multi-scale semantic features of the point cloud, we perform the pooling operation on different receptive fields. Specifically, we evenly divide the per-point features into 1, 2, 4, 8, 16, and 32 portions, respectively. Each portion of features represents a different receptive field. We apply max pooling on each portion to obtain features of different scales. Then feed these features into a Multilayer Perceptron (MLP) to obtain the point cloud feature with multi-scale information F_s and then concatenate it with the global feature F_g based on PointNet [21] to generate the required conditional feature c.

Conditional Feature Fusion module: The Conditional Feature Fusion (CFF) module is designed to propagate the conditional feature c obtained from the aforementioned CFE module in the contour generation process, as shown in Fig. 4(a). In this module, the conditional feature c is applied to each time step of the generation process. Since the generation results depend on the time step t, we introduce a time embedding vector composed of $[\beta_t, \sin(\beta_t), \cos(\beta_t)]$ to ensure the condition at time t. The inputs of this module are the conditional feature c, the point cloud at time t in the generation, and the parameter β_t regarding time t. We use a quaternions $c = [\beta_t, \sin(\beta_t), \cos(\beta_t), c]$ as the context embedding vector. Then, quaternion c is inserted into each layer of the generation process. The feature information c is aggregated to point cloud X^t, and the feature is eventually propagated to each step of the denoised point cloud. The CFF module consists of a series of ConcatSquashLinear layers and spatial attention layers [22]. In the dimensionality-raising stage of feature fusion, spatial attention layers are introduced to emphasize the edge information of the point cloud. The conditional feature and the point cloud at time t are fused in different feature dimensions. The details of ConcatSquashLinear are shown in Fig. 4(b). The conditional feature c is mapped to the range (0,1) by a sigmoid function. Then, it is multiplied with the transformed point cloud X^t and added back to feature c.

IV. Experiment

A. Datasets and Settings

Our network is trained on a ShapeNet-based synthetic dataset and the publicly available ABC dataset. The ShapeNet dataset [23] is a public dataset containing 55 classes and 513,000 CAD models, but it does not provide ground truth of point cloud contours. We select a subset of shapes from the ShapeNet dataset and label their contours, thus creating a synthetic dataset with contour labels, namely the ShapeNet-synthetic dataset. The ABC dataset [24] is a public dataset that contains over 1 million CAD models of machine parts with the ground truth of corresponding contours. We select 2000 pairs of point clouds, including complete objects and corresponding contours for network training. All the datasets used in our experiments were randomly divided into training and testing sets with a ratio of 90% for training and 10% for testing. In our experiments, we set the number of points to 2048 for complete point clouds and 1024 for contour point clouds. The network is trained on Ubuntu20.04 and NVIDIA GeForce RTX 2080 Ti GPU with the environment of CUDA 10.1 and PyTorch 1.11.0. In the training, the learning rate is set as 10^{-3}, and the number of generation steps is 100.

B. Results of Contour Generation

The visualization results of contour generation on the ShapeNet-synthetic dataset by our method are shown in Fig. 5. Luo et al. [10] provide an effective method for point cloud generation via the diffusion model, so we select it as the comparative benchmark method. As shown in the figure, our

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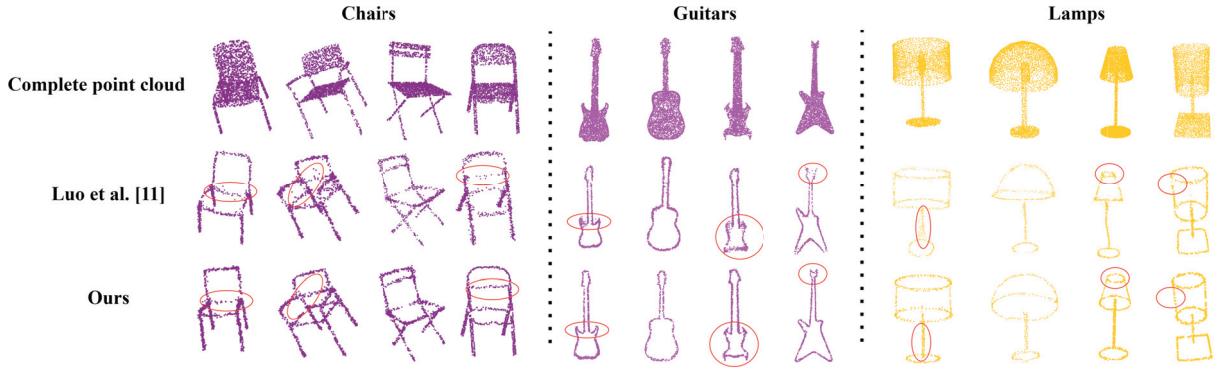

Fig. 5. The visualization results and comparisons on the ShapNet-synthetic dataset.

TABLE I
COMPARISON OF POINT CLOUD CONTOUR GENERATION PERFORMANCE. CD IS MULTIPLIED BY 10^2 , EMD IS MULTIPLIED BY 10^1.

	Chair		Guitar		Plane		Lamp	
Model	CD	EMD	CD	EMD	CD	EMD	CD	EMD
Luo et al. [10]	0.82	1.13	**0.18**	0.95	0.40	0.85	3.41	3.46
Pointflow [8]	0.83	1.36	0.93	1.08	4.21	1.82	**1.83**	**1.61**
Snowflakenet [18]	1.26	2.21	0.28	2.01	1.25	0.93	5.13	4.81
Ours	**0.64**	**0.99**	0.20	**0.53**	**0.24**	**0.66**	2.38	1.67

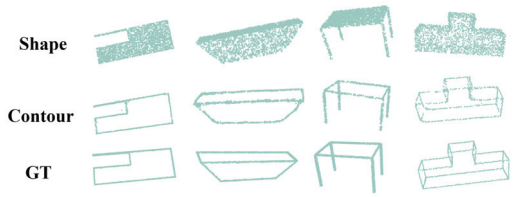

Fig. 6. Some representative examples of contours generation by our method on the ABC dataset.

method effectively generates the contour of point cloud shapes, resulting in a more uniform and continuous distribution of contour points with fewer outliers. Fig. 6 shows the contour generation results on the ABC dataset. The first row is the dataset's original point clouds of machine parts. The second row is the corresponding contours. Ground truth provided by the dataset is listed in the third row for comparison to demonstrate the effectiveness of our method.

Moreover, we use the Chamfer Distance (CD) and Earth Mover's Distance (EMD), two extensively employed metrics in the shape similarity assessment [8], to evaluate the contour generation quality on the ShapeNet-synthetic dataset. Some state-of-the-art point cloud generation works are selected for generating 3D contour as the comparative methods, including Luo et al. [10], Pointflow [8], and Snowflakenet [18]. The quantitative results are reported in Table I. Our method achieves the best performance in the categories of "chair" and "plane" in both metrics. For guitars, we excelled in the EMD metric and closely approached the best result in the CD metric. In the case of "lamp" where PointFlow [8] reaches the best performance, our results are very close to theirs.

C. Generalization of the network

The generalization capability of our method for various types of point clouds is extensively discussed in this section. We train our network on the ShapeNet-synthetic dataset and generate contours for the corresponding categories in the ModelNet40 dataset [25]. The ModelNet40 dataset [25] is also a well-known dataset for point cloud processing. It shares some common categories with the ShapeNet [23] dataset, but the 3D models have different shapes. Therefore, for the pre-trained network, the shapes on the ModelNet40 dataset are unseen. Fig. 7 shows two categories of shapes in the ModelNet40 dataset and their corresponding contours generated by our method. It can be observed that our proposed network has outstanding generalization capability, achieving accurate and continuous contour generation in cross-dataset inference.

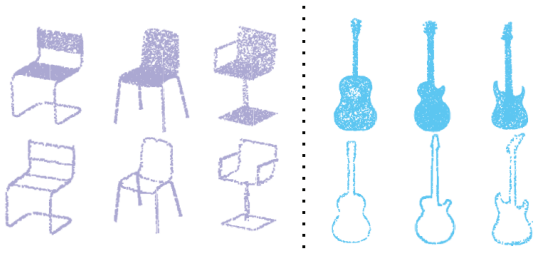

Fig. 7. Some representative examples of contours generation by our pre-trained network on ModelNet40 dataset

D. Results of Contour Classification

An accurate contour of an object should effectively reflect its shape. Therefore, we additionally use the generated contours for the shape classification task to validate the effectiveness of our method. We compared the classification accuracy

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of the generated contours and complete point clouds for four categories using two classification networks, DGCNN [20] and PointNet [21]. It is worth noting that the classification accuracy of the contour is obtained using a pre-trained classifier on the complete point cloud (1024 points). The classification accuracy obtained from contours is very close to the accuracy obtained from the complete point cloud, as reported in Table II, which indicates the generated contours effectively represent the shape of the original point cloud.

TABLE II
COMPARISON OF CLASSIFICATION ACCURACY BETWEEN CONTOURS AND COMPLETE POINT CLOUDS. THE CONTOURS* IS OBTAINED USING THE CLASSIFIER PRE-TRAINED ON THE COMPLETE POINT CLOUD.

Classifier	**DGCNN** [20]		**PointNet** [21]	
	contour*	complete	contour*	complete
Chair	0.87	0.96	0.90	0.96
Guitar	0.93	1	0.87	0.90
Plane	0.91	1	0.85	1
Lamp	0.88	1	0.89	1

V. CONCLUSION

The contour of a 3D object plays a crucial role in object recognition and localization. We propose a diffusion-based contour generation network that generates corresponding contours from noise under the guidance of complete point clouds. In the network, a CFE module is designed for extracting multi-scale point cloud conditional features, and a CFF module is designed to fuse the obtained conditional feature in the denoising generation process as guidance. We conduct experiments on public and synthetic datasets to qualitatively and quantitatively validate the effectiveness of our method. Furthermore, we test the pre-trained network on cross-datasets, demonstrating the good generalization ability of our method. In the future, we expect to extend the proposed method to the field of scene generation.

VI. ACKNOWLEDGEMENT

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